

WhatsApp Support Gateway - Video Demo Teleprompter

Dual-Pane Showcase: Sub-30ms Decoupling, Redis Streams, Anti-IDOR Security & Live WebSocket Takeover

PRE-RECORDING QUICK CHECKLIST:

1. Terminal: `uv run uvicorn gateway.main:app --reload --port 8000`
2. Browser: Open `http://localhost:8000/demo` (100% zoom, both panes visible)
3. Screen Recorder: Press **Print Screen** key (or open Loom/OBS)

STEP 0: INTRO & ARCHITECTURE OVERVIEW

ACTION (WHAT TO CLICK/TYPE):

Hover mouse over WhatsApp phone on left, then over real-time metrics on right.

ON SCREEN (VISUAL REACTION):

Browser at `http://localhost:8000/demo` showing phone simulator and telemetry console.

SPOKEN SCRIPT (WHAT TO SAY):

"Hi there! Today I'm demonstrating an enterprise-grade WhatsApp Business API Support Gateway built with Python, FastAPI, and Meta Cloud API v20.

Most WhatsApp chatbots fail in production because AI calls take over 3 seconds, triggering Meta's retry storm and spamming customers with duplicate messages. Here on the left is our WhatsApp phone simulator, and on the right is our real-time telemetry cockpit showing sub-30ms webhook decoupling, HMAC verification, distributed Redis Streams, and live human takeover."

STEP 1: ORDER TRACKING (OWNER - SUB-30MS & READ RECEIPTS)

ACTION (WHAT TO CLICK/TYPE):

Click chip: **1. Track #ORD-1001**

ON SCREEN (VISUAL REACTION):

3 green typing dots appear ('typing...'), double grey ticks turn bright blue ('✓✓' read receipt), latency shows <20ms, bot replies with FedEx status and buttons.

SPOKEN SCRIPT (WHAT TO SAY):

"First, let's track an order. Notice the natural language processing: it recognized lowercase 'ord 1001' and mapped it instantly.

Look at the telemetry: the webhook acknowledged the message in under 20 milliseconds, verified the HMAC-SHA256 signature, dispatched an official Meta 'mark_read' receipt turning the ticks blue, rendered realistic typing dots, and retrieved the order record with interactive action buttons."

STEP 2: MULTI-TURN CONTEXT & RETURN POLICY ENFORCEMENT

ACTION (WHAT TO CLICK/TYPE):

Click chip: 2. Can I return it?

ON SCREEN (VISUAL REACTION):

Bot resolves 'it' to #ORD-1001 and replies that in-transit orders cannot be returned until delivered.

SPOKEN SCRIPT (WHAT TO SAY):

"Next is multi-turn contextual memory and business policy enforcement. Notice I didn't re-type the order number; I simply asked 'Can I return it?'. The engine automatically resolved the pronoun 'it' to order #ORD-1001.

Because the package is still in transit with FedEx, the agent strictly enforces return policy and explains that returns are only available once the item has been delivered."

STEP 3: ANTI-IDOR SECURITY VERIFICATION CHALLENGE

ACTION (WHAT TO CLICK/TYPE):

Click chip: 3. Track #ORD-1002 (Security Check)

ON SCREEN (VISUAL REACTION):

Bot intercepts query for #ORD-1002 (belongs to another phone), withholds shipment details, and asks for the last 4 digits on file.

SPOKEN SCRIPT (WHAT TO SAY):

"Now let's see our Anti-IDOR privacy protection. When a caller queries order #ORD-1002—which belongs to a different customer phone number—the gateway immediately blocks access.

It withholds all private shipment information and issues a security challenge requiring the caller to verify the last 4 digits on file."

STEP 4: OWNERSHIP AUTHENTICATION

ACTION (WHAT TO CLICK/TYPE):

Click chip: 4. Verify: 6543

ON SCREEN (VISUAL REACTION):

Bot confirms 'Security Verification Successful!' and unlocks #ORD-1002 status ('PROCESSING').

SPOKEN SCRIPT (WHAT TO SAY):

"Once the customer enters the correct digits '6543', the security layer confirms ownership, unlocks the session, and presents the order status."

STEP 5: DELIVERED ORDER RETURN REQUEST

ACTION (WHAT TO CLICK/TYPE):

Click chip: 5. Return #ORD-1003

ON SCREEN (VISUAL REACTION):

Bot detects delivered order #ORD-1003 belongs to another phone (...2233) and requests 4-digit verification for return approval.

SPOKEN SCRIPT (WHAT TO SAY):

"Now let's request a return on a delivered order, #ORD-1003. Again, because it's associated with a separate account, the gateway demands verification before issuing prepaid postage."

STEP 6: PROGRAMMATIC VECTOR PDF RETURN LABEL & BARCODE

ACTION (WHAT TO CLICK/TYPE):

Click chip: 6. Verify: 2233 (Get PDF)

Then click the PDF Document Card in the phone.

ON SCREEN (VISUAL REACTION):

Bot verifies ownership, approves return, and sends document card: ■ return_label_ORD-1003.pdf. Clicking opens vector PDF in new tab with FedEx label, Code128 barcode & RMA packing slip.

SPOKEN SCRIPT (WHAT TO SAY):

"When I verify with '2233', the return is immediately approved."

The gateway doesn't just send text—it programmatically generates a vector PDF shipping label using ReportLab. Clicking the document card opens the live label, complete with prepaid FedEx Ground routing, a scannable Code128 barcode encoding the RMA number, and an itemized warehouse packing slip."

STEP 7: SHOPIFY ADMIN REST API ADAPTER

ACTION (WHAT TO CLICK/TYPE):

(Switch back to demo tab) Click chip: 7. Shopify #1001

ON SCREEN (VISUAL REACTION):

Telemetry shows routing through ShopifyOrderAdapter. Bot returns live line items ('Classic Leather Jacket') and Shopify fulfillment status.

SPOKEN SCRIPT (WHAT TO SAY):

"The system is built on clean architectural ports and adapters. With a single environment variable, we can switch from our local database to a live Shopify Admin REST API. Here you see it parsing Shopify line items, fulfillment stages, and tracking numbers seamlessly."

STEP 8: HUMAN ESCALATION & STRICT BOT MUTING

ACTION (WHAT TO CLICK/TYPE):

Click chip: 8. Talk to Human

ON SCREEN (VISUAL REACTION):

Right pane: Session State switches to amber **ESCALATED_HUMAN**. Outbound webhook fires to n8n/Zendesk. Operator Card expands showing '■ Live WebSocket Connected'.

SPOKEN SCRIPT (WHAT TO SAY):

"Now for the most critical enterprise capability: Stateful Human Escalation and Live Operator Takeover."

*When the customer requests human help, an external webhook fires to Zendesk or n8n, and strict bot muting engages. Notice the session state is now **ESCALATED_HUMAN**, and our Operator Station has opened a real-time WebSocket bridge."*

STEP 9: BOT MUTING VERIFICATION (ZERO SPAM)

ACTION (WHAT TO CLICK/TYPE):

In phone chat input, type: **Where is my refund?** and click send.

ON SCREEN (VISUAL REACTION):

Message appears in WhatsApp feed. **Bot stays completely silent.** Customer message appears live inside Operator Chat Feed on right.

SPOKEN SCRIPT (WHAT TO SAY):

"Watch what happens when the customer types 'Where is my refund?': the bot stays completely silent so it never talks over human staff. Instead, the message was broadcast across the WebSocket and appeared directly in the operator console."

STEP 10: REAL-TIME 2-WAY OPERATOR CHAT TAKEOVER

ACTION (WHAT TO CLICK/TYPE):

In Operator Desk input, type: **Hi, Agent Sarah here! Your refund of \$189.50 has been released.** and click **Send as Agent**.

ON SCREEN (VISUAL REACTION):

Message appears immediately on customer phone with blue ■ Agent Sarah (Live Support) badge. Telemetry confirms outbound WhatsApp dispatch and transcript logging.

SPOKEN SCRIPT (WHAT TO SAY):

"As the human agent, I type: 'Hi, Agent Sarah here! Your refund has been released'."

The moment I click Send, it dispatches over the WhatsApp channel, pops up immediately on the customer's phone, and is saved into the permanent customer transcript."

STEP 11: SESSION RESOLUTION

ACTION (WHAT TO CLICK/TYPE):

Click button: Unmute Bot / Resolve

ON SCREEN (VISUAL REACTION):

Operator Station closes. Session State returns to green **ACTIVE_BOT**. Bot status resets to '● Online (Bot Active)'.

SPOKEN SCRIPT (WHAT TO SAY):

"Once the customer inquiry is resolved, clicking 'Unmute Bot' seamlessly re-activates automated bot handling for future inquiries."

STEP 12: PRODUCTION RIGOR & 94-TEST SUITE

ACTION (WHAT TO CLICK/TYPE):

Switch to Terminal window and run: `uv run pytest`

ON SCREEN (VISUAL REACTION):

All 94 unit & integration tests pass green in under 1.5 seconds.

SPOKEN SCRIPT (WHAT TO SAY):

"Under the hood, this is a fully tested production system. The application is containerized with Docker Compose and Redis Streams for horizontal worker scaling."

The entire codebase features strict mypy typing, ruff linting, and 94 automated unit and integration tests passing in under 1.5 seconds."

Thank you for watching, and I look forward to bringing this high-reliability architecture to your customer support operations!"